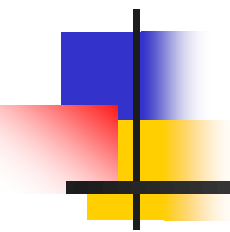


# 04. Java Class – a new concept



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Class fundamentals

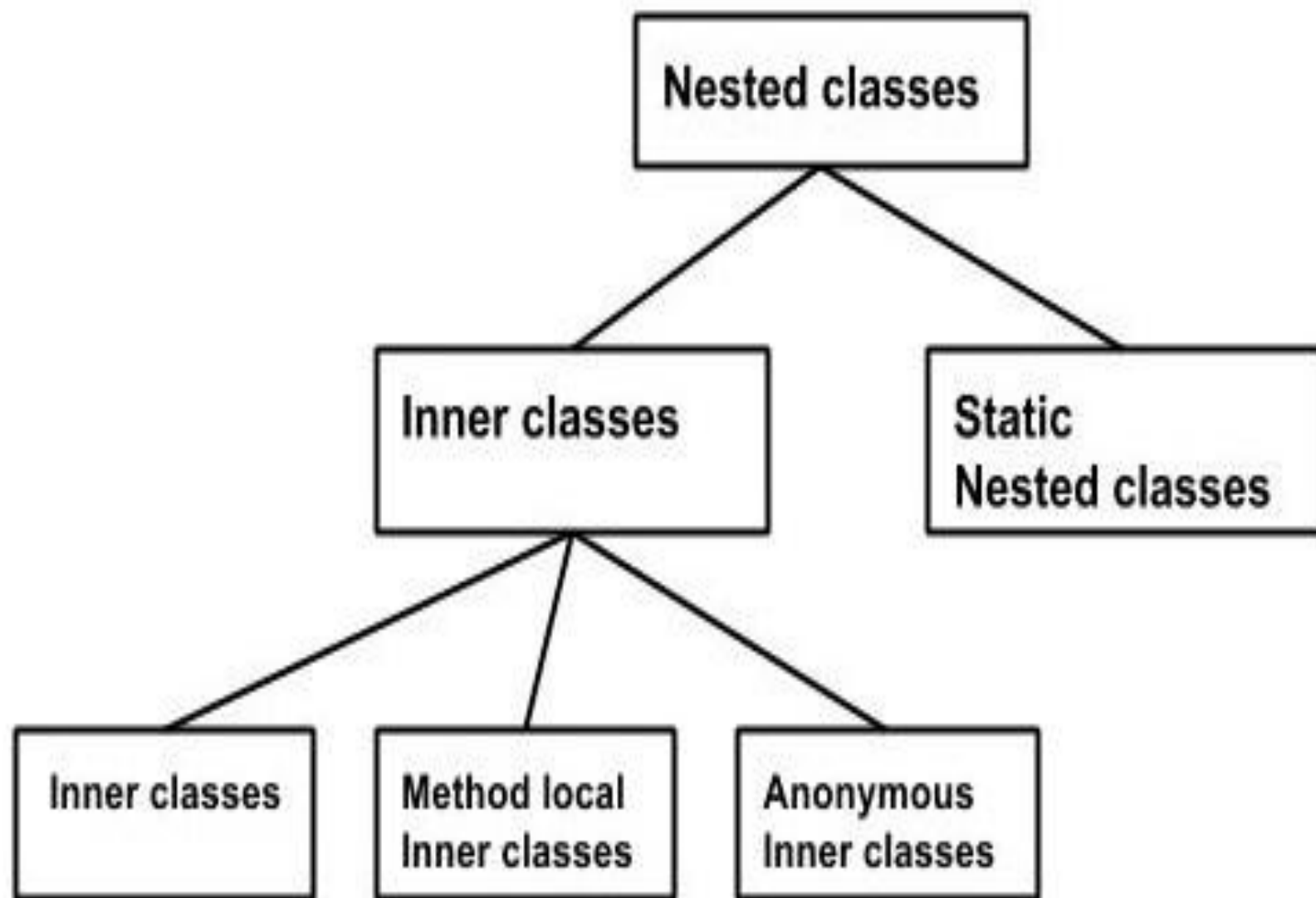
Methods

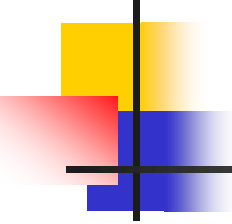
Constructors

Garbage Collection – Java Way

Access Modifiers

Inner Classes



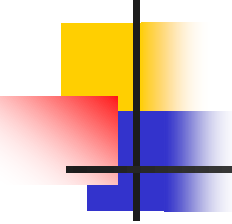


# Nested Classes

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- The Java programming language allows you to define a class within another class. Such a class is called a *nested class*.

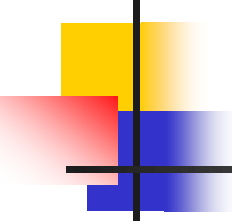
```
class OuterClass {  
    ...  
    class NestedClass {  
        ...  
    }  
}
```



# Nested Classes

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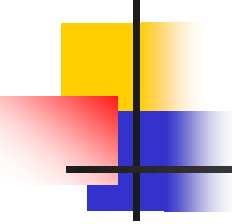
- Nested classes are divided into two categories: static and non-static. Nested classes that are declared static are simply called *static nested classes*. Non-static nested classes are called *inner classes*



# Nested Classes

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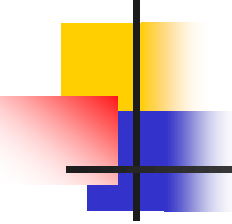
```
class OuterClass {  
    ...  
    static class StaticNestedClass {  
        ...  
    }  
    class InnerClass {  
        ...  
    }  
}
```



# Static Nested Classes

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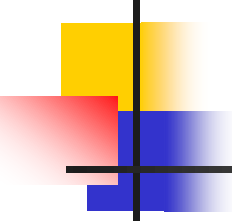
- As with class methods and variables, a static nested class is associated with its outer class. And like static class methods, a static nested class cannot refer directly to instance variables or methods defined in its enclosing class — it can use them only through an object reference.



# Inner Classes

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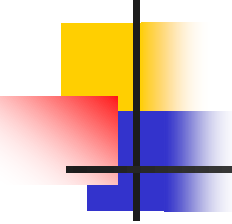
- As with instance methods and variables, an inner class is associated with an instance of its enclosing class and has direct access to that object's methods and fields. Also, because an inner class is associated with an instance, it cannot define any static members itself.



# Local and Anonymous Inner Classes

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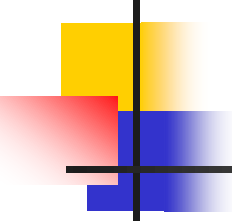
- There are two additional types of inner classes. You can declare an inner class within the body of a method. Such a class is known as a *local inner class*. You can also declare an inner class within the body of a method without naming it. These classes are known as *anonymous inner classes*.



# Benefits of Inner Class

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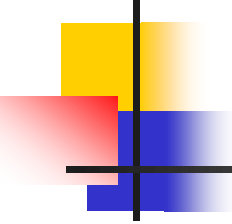
- If a class is useful to only one class, it makes sense to keep it nested and together. It helps in the packaging of the classes.
- Java inner classes implements encapsulation. Note that inner classes can access outer class private members and at the same time we can hide inner class from outer world.



# Benefits of Inner Class

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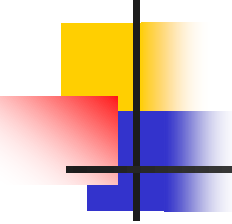
- Keeping the small class within top-level classes places the code closer to where it is used and makes the code more readable and maintainable.



# Features - **Keywords**

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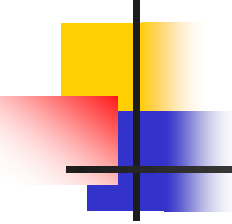
- Static Nested Class
  - Static
- Member Inner Class
  - Non-static
- Local Inner Class
  - Defined inside a method or block
- Anonymous Inner Class
  - Defined inline, no name



# Features - **Access to Outer Class**

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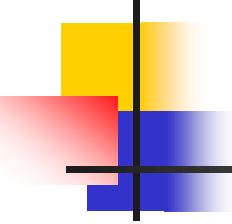
- Static Nested Class
  - Only static members
- Member Inner Class
  - Both static and non-static members



# Features - **Access to Outer Class**

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- Local Inner Class
  - Can access final or effectively final local variables and outer class members
- Anonymous Inner Class
  - Can access outer class and method variables



# Features - **Instantiation**

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- Static Nested Class
  - Without an instance of the outer class
- Member Inner Class
  - Requires an instance of the outer class
- Local Inner Class
  - Limited to the scope where it is defined
- Anonymous Inner Class
  - Instantiated at the point of creation